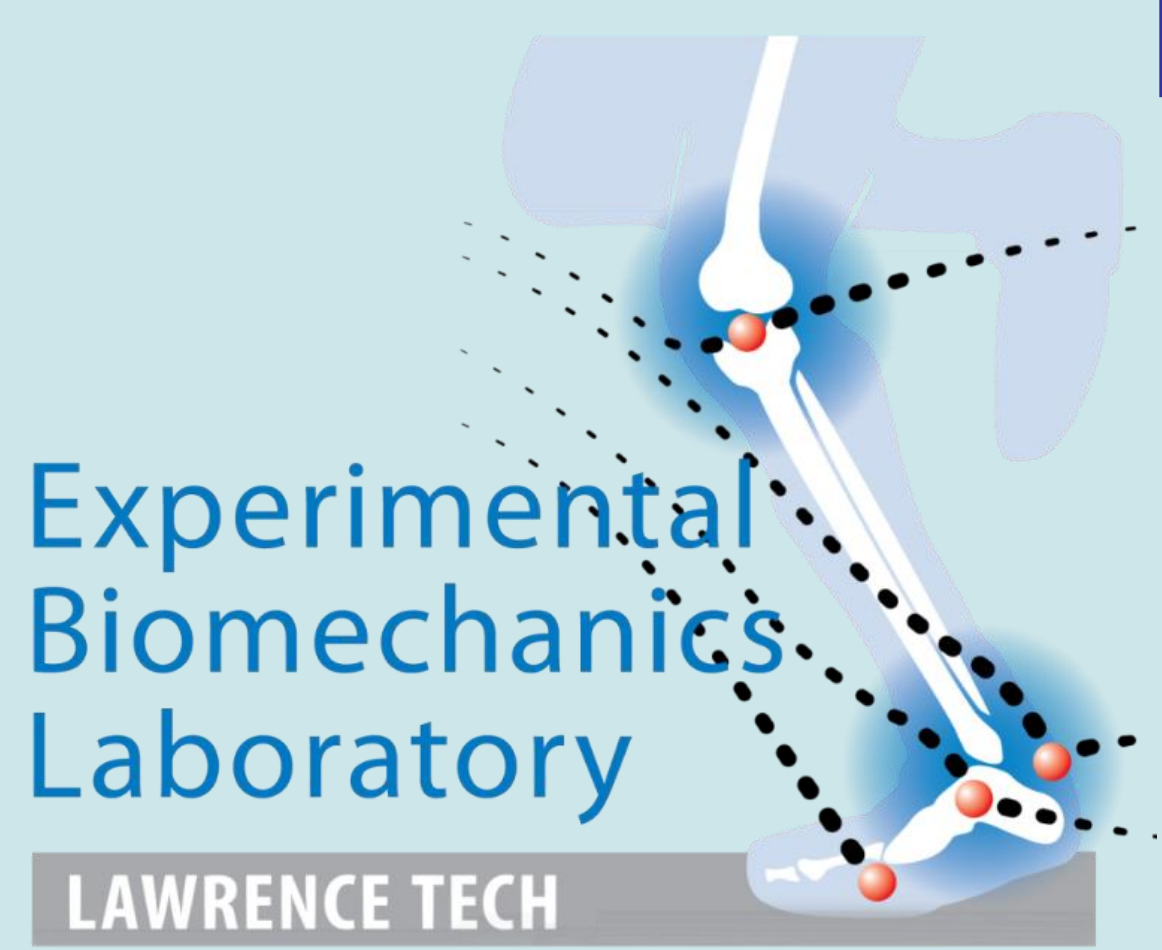




Practicing Design and Prototyping in Biomedical Engineering Programs

Eric G. Meyer and Mansoor Nasir

Biomedical Engineering, Lawrence Technological University

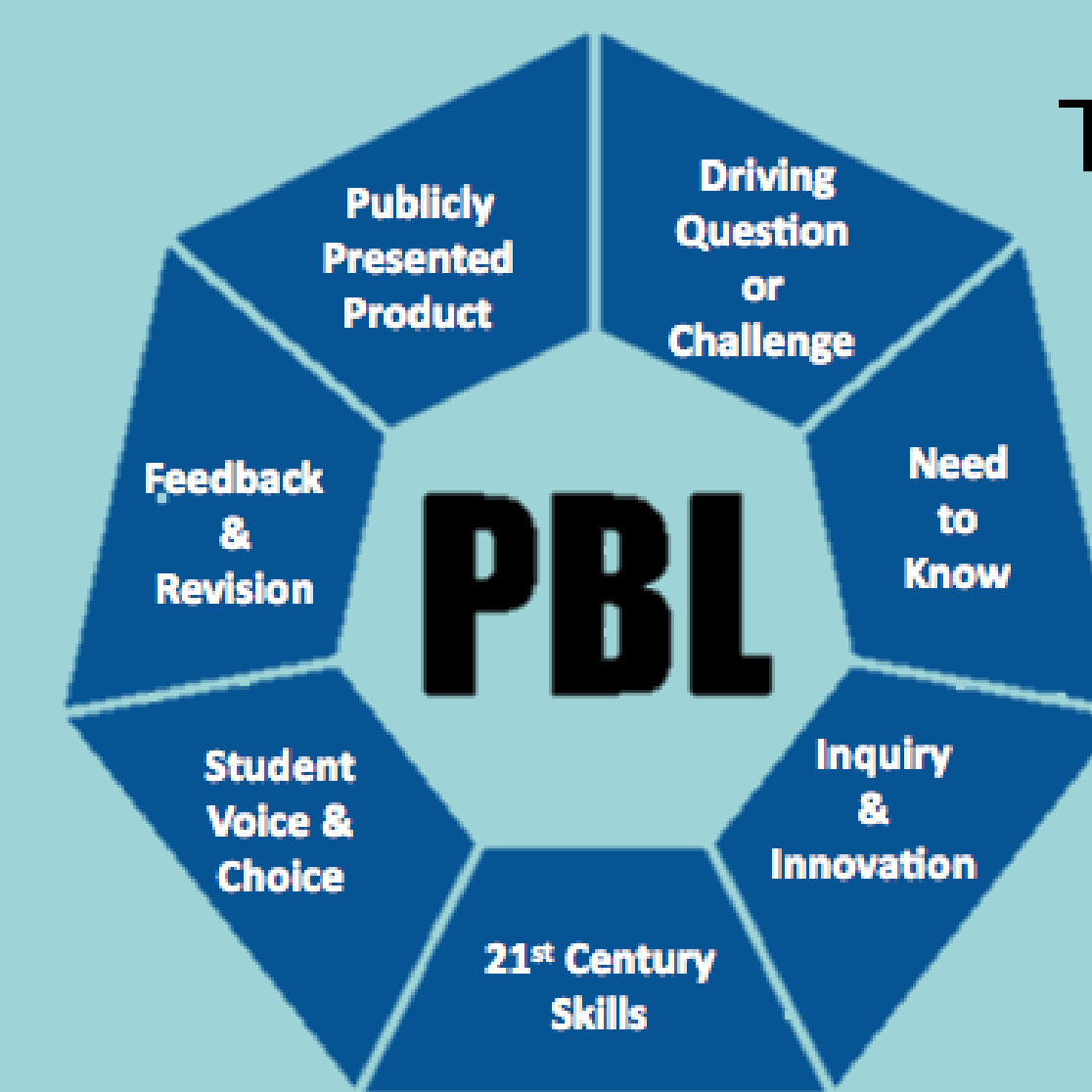
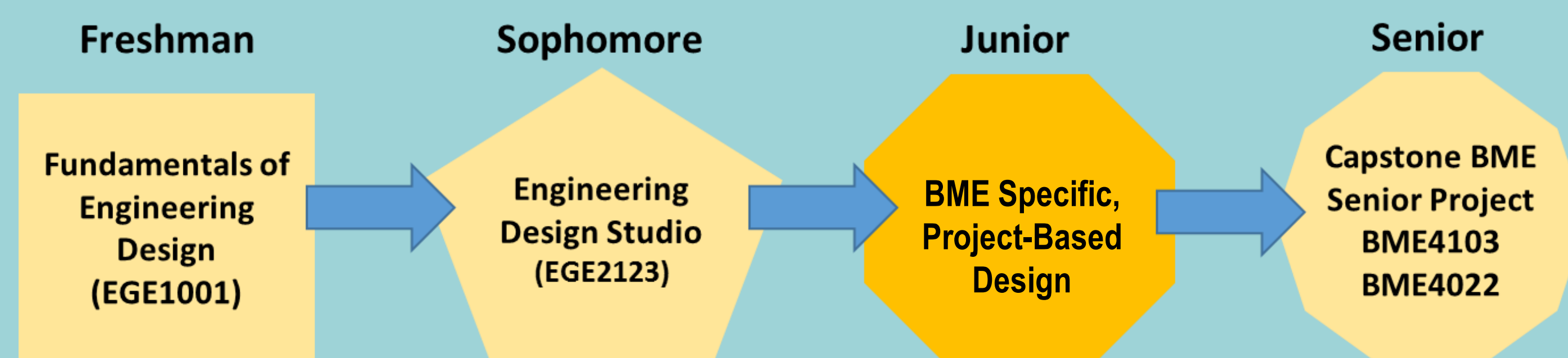
**BIOSENSORS
LAB**

Lawrence Tech

Engineering Design in BME

Biomedical engineering curriculum provides students with repeated design-implement experiences.

Inductive learning activities, like “Project Based” (PBL) and “Entrepreneurial Minded” (EML) cover fundamental science content, engineering principles and analytical capabilities in a real-world and authentic context. The aim is to develop students who are better at adapting to new trends, embracing creativity and leadership, understanding engineering impacts on society and business, as well as providing more opportunities to experience engineering design beyond the capstone projects. The Junior design courses increase students’ topic specific capabilities and confidence to prepare them for their Senior Project and engineering careers.



**Entrepreneurial
Minded Learning**
Curiosity
Connections
Creating Value

This is what is real world problems look like

Problems / Needs
↓
Many Constraints
↓
Many Tools
↓
Many Solutions



Wearable Technology Studio

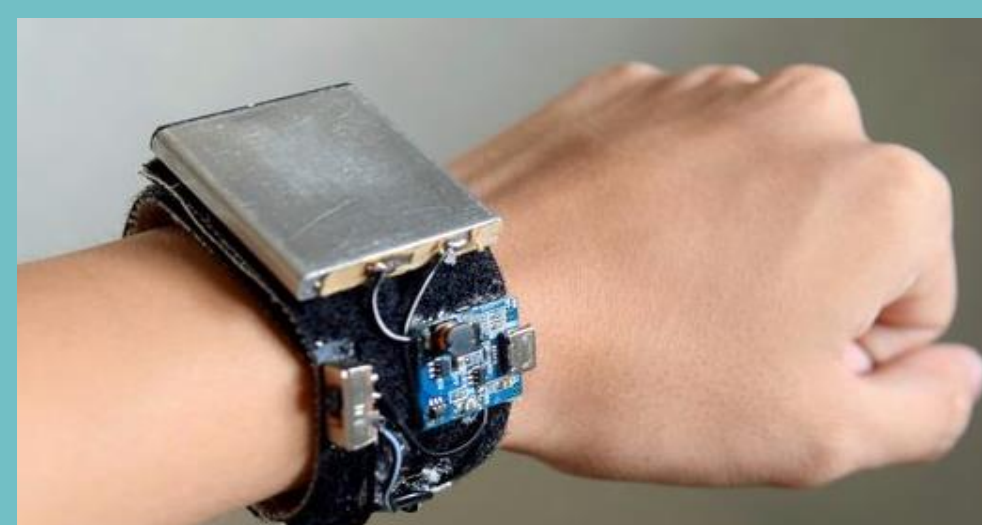
A studio format, product and engineering design course introducing various aspects of wearable technology.

Wearable technology is reducing the barriers for companies and individuals to create new products for fashion, sports, lifestyle, computing, and health industries. This course focuses on skill building, such as; soldering, sewing, CAD, 3D printing and machining techniques, building with Arduino, using sensors and e-textiles. Practical information of relevant topics such as MEMS sensors, mobile apps programming, user interaction, materials and manufacturability, and marketing are also covered. Three group projects promote creativity and hands on experience creating prototypes and designing wearable products.

Project 1
LED Slipper



Project 2
Fitness Band/Smart Watch



Project 3
Product Innovation



Semester Schedule

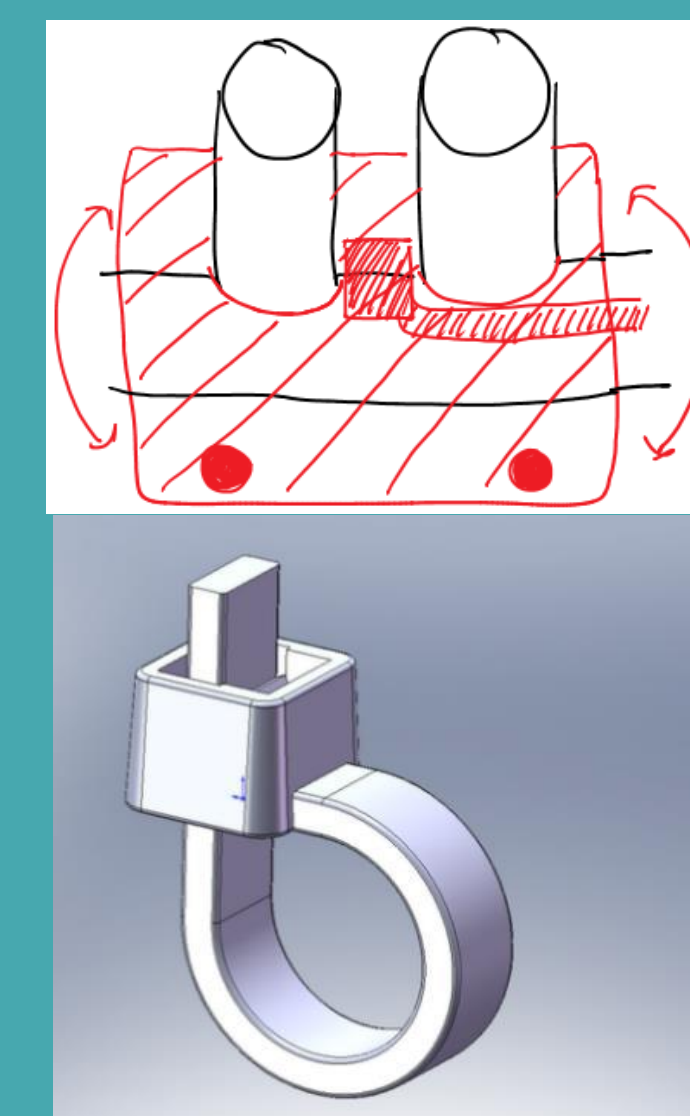
Week	Topic (30 min)	In Class Activity (60 min)	Project (120 min)
1	Introduction	Virtual/Augmented Reality	0.) Reverse Eng.
2	Textiles	Cutting, stitching, sewing	1.) E-Slipper Intro Circuit design Pressure Switch
3	Basic Electronics	Soldering	
4	Soft Circuits	Electronic textiles	
5	Microcontrollers	Arduino / Lilypad	2.) Fitness Band Intro Step Counter CAD Case Fabrication
6	Sensor and Actuator	Temperature / Pedometer	
7	Digital Design	3D Scanning to STL files	
8	Digital Fabrication	Laser Cutting and CAM	3.) Open Ended Intro Usability Design Software Design Communications Sustainability Fit and Finish Marketing Pitch
9	Social Aspects of Design	Design thinking	
10	User Interface Design	Unity programming	
11	Software Design	Leap and Unreal	
12	Data Processing	Using Big Data	
13	Privacy and Security	Fair Information Practices	
14	Manufacturing	Commercialization	
15	Product Design	Business Model Generation	
FE	Final Project Presentations		"Kickstarter" page and prototype

Medical Device Design

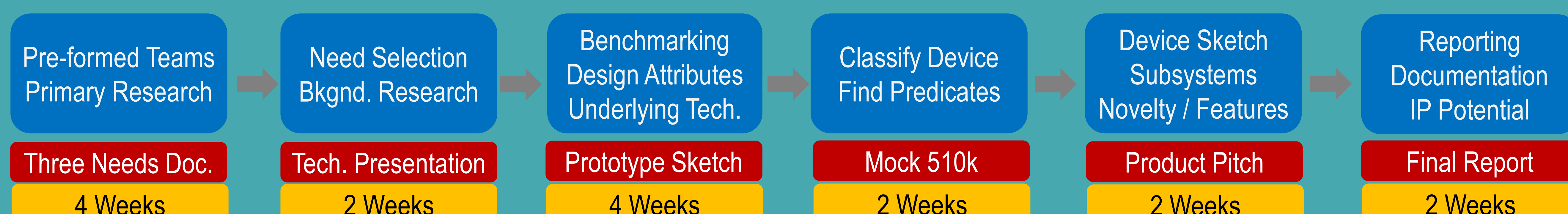
A soft introduction to senior capstone engineering design process focusing on the principles and design of medical devices.

Medical devices are advanced engineering systems. This course focuses on the working principles, design, manufacture, reliability and some regulatory hurdles involved in the development of biomedical devices and sensors. Case study devices such as ECG monitors, pacemakers, blood pressure monitors and glucose sensors are explored in detail. A course project starts with ideation and follows the design process for medical devices (Benchmarking, Attributes, Concept, Communication).

Design Prototypes



Semester Schedule



Moving Forward

- How effective is it? → Assessment
- Finding a skill balance → Technical vs Entrepreneurial
- Uniformity in student experience → Course Comparison

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We acknowledge contributions to the Wearable Technology Studio course from LTU faculty: Bryce Evans, Game Art; Justin Ku, Management; Dan Shargel, Humanities; Franco Delogu, Psychology

References [1] Froyd et al. IEEE 2012. [2] Crawley et al. Springer 2014. [3] Fairweather J. National Research Council, 2008. [4] Meyer and Nasir. ASEE 2015. [5] Gerhart et al. ASEE 2013. [6] Bell-Huff et al. ASEE 2016.